Lifetime Performance and Degradation of Floating Photovoltaic Systems Under Humid and Saline Conditions

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***Abstract*—This study develops a coupled hourly simulation framework to quantify the competing thermal, humidity, soiling, potential-induced degradation (PID) and chloride-corrosion effects on floating photovoltaic (FPV) systems in hot, humid environments. A single over-water microclimate drives module temperature, module-surface humidity and dew, a dust and sea-salt soiling balance with NaCl deliquescence, and a four-mechanism module degradation model. The mechanisms are intrinsic ageing, Peck temperature–humidity damage with lagged encapsulant moisture, PID with salt-assisted surface leakage, and corrosion scaled by time of wetness and chloride deposition. For a representative Visakhapatnam (India) climate, the model compares coastal land PV, freshwater FPV and saline FPV. With the baseline parameters, the model predicts baseline year-1 yield differences relative to land of $+1.6\%$ (freshwater) and -1.1% (saline), and baseline 25-year energy differences of $+2.1\%$ and -4.3%; the predicted saline degradation rate is 1.16%/yr against 0.88%/yr on land. A 1,000-sample Monte Carlo over 28 uncertain parameters, reported separately from weather variability, gives median 25-year differences of $+1.5\%$ (95% interval $[-3.6, 6.3]\%$) for freshwater FPV and -5.1% ($[-11.5, 0.7]\%$) for saline FPV. The model therefore predicts a lifetime penalty for saline FPV across most of the plausible parameter space. The freshwater benefit is conditional: it is positive in 75% of samples, depends strongly on the thermal characteristics of the floating structure, and in this model vanishes if the dew point over the water rises by more than about 2.7 K. Monte Carlo convergence is verified with 5,000 samples. Mitigation is modelled with explicitly assumed reduction factors, not inferred from certification. All results are model-based scenario predictions that still require field validation.**

***Index Terms*—Floating photovoltaics, humidity, salinity, soiling, potential-induced degradation, corrosion, module temperature, uncertainty quantification, degradation rate, levelized cost of energy.**

NOMENCLATURE

G	Plane-of-array (POA) irradiance (W/m^2).
T_a, T_d, T_m	Air, dew-point and module temperature (°C).
v	Wind speed (m/s).
U_0, U_1	Faiman heat-loss coefficients.
$RH_{\mathrm{s}}, RH_{\mathrm{e}}$	Module-surface and encapsulant relative humidity (%).
m_d, m_s	Dust and salt surface density (g/m^2).
SR	Soiling ratio.
TOW	Time of wetness (h/yr).
D_{Cl}	Chloride deposition rate (mg/(m^2 d)).
R	Annual module power degradation rate (%/yr).
$\mu_{\mathrm{PID}}, \mu_{\mathrm{C}}$	Assumed mitigation factors for PID and corrosion.
E_a	Activation energy (eV).
k_B	Boltzmann constant (8.617×10^{-5} eV/K).
PR	Performance ratio.

Manuscript prepared October 2026; revised after peer review.

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The simulation code, input parameters, random seeds and all scripts used to produce the results are openly available (see the Data Availability section).

I. INTRODUCTION

FLOATING photovoltaic (FPV) systems place PV arrays on pontoons over reservoirs, lakes, lagoons and sheltered coastal waters. They avoid competition for land, reduce evaporation from water-supply reservoirs and can share grid infrastructure with hydropower plants [1]–[5]. India has many reservoirs, salt pans and backwaters along a coast with high irradiation, so FPV is attractive there, but many candidate sites are hot, humid and exposed to marine aerosol [6].

The performance case for FPV usually rests on the cooling effect of the water. In tropical Singapore, well-ventilated FPV designs ran cooler and reached higher performance ratios than a rooftop reference [7], with irradiance-weighted temperatures up to 14.5 °C lower [8]. Offshore simulations also predict higher yields than on land [9]. More recent measurements show that the cooling effect varies widely with design. Modules that are not in thermal contact with the water behave much like land-based modules [10]. Measured heat-loss coefficients span 22.6–33.7 W/(m^2 K) for several air-cooled systems [11], [12], and wind-dependent terms of 2.7–5.1 W s/(m^3 K) have been reported [13]. A multi-year, humid-subtropical dataset found the FPV array only about 0.8 K cooler than a co-located land reference [14].

The water body also raises humidity and, at saline sites, airborne salt. The Peck model describes humidity-driven damage as an Arrhenius function of temperature times a power of relative humidity [15], and moisture enters polymer-backsheet modules over hours to days [16]. Combined-stressor lifetime models confirm that humidity is a first-order variable [17]. PID is strongly accelerated by humidity [18]–[22]. Sea salt lowers the surface resistance of module glass by several orders of magnitude as RH rises [23], and salt raises leakage current [24]. Field surveys report higher degradation in hot climates [25], and in India the hot–humid zones are among the worst affected [26]. Chloride deposition and time of wetness

TABLE I
SCOPE OF REPRESENTATIVE STUDIES (✓ = ADDRESSED)

Study	FPV	Therm.	Hum.	Soil.	Salt	PID	Corr.	Life	LCOE
Liu *et al.* [7]	✓	✓							
Dörenkämper *et al.* [8]	✓	✓							
Lindholm *et al.* [10]	✓	✓							
Golroodbari, van Sark [9]	✓	✓							
Ahmad *et al.* [29]	✓	✓		✓	✓				
Kaaya *et al.* [17]		✓	✓					✓	
Mannino *et al.* [31]	✓	✓	✓					✓	
Ajmal *et al.* [32]	✓	✓	✓		✓			✓	
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓

define atmospheric corrosivity [27], and salt mist is a module qualification stress [28]. Salt crystallization dominates the deposits on coastal FPV modules [29], and salt exposure acts on PV through soiling, corrosion, leakage-driven PID and barrier ageing [30].

Two recent studies come closest to this work. Mannino *et al.* [31] applied existing degradation models to onshore and offshore meteorological data. Ajmal *et al.* [32] added salinity to climatic degradation models for offshore modules and calibrated them by optimization. Table I compares the scope of representative studies. Among the studies reviewed here, none was identified that resolves, within one consistent microclimate, both the yield benefits of FPV and the humidity-, salt- and PID-driven penalties, and that compares freshwater with saline sites over the plant lifetime and in cost terms.

This paper makes the following contributions:

1) A coupled hourly model chain in which one over-water microclimate drives module temperature, dew, spectral response, soiling and four module-degradation mechanisms, so that their trade-offs are resolved consistently.
2) A moisture formulation based on module-surface RH with a lagged encapsulant response. It captures why cooler modules can see more moisture, and it couples PID to salt-assisted surface leakage.
3) A comparison of coastal land PV, freshwater FPV and saline FPV covering year-1 losses, mechanism-resolved degradation, 25-year energy and LCOE. Mitigation options are evaluated one by one, with explicitly assumed effects.
4) An uncertainty analysis that separates weather variability (20 synthetic years) from parameter uncertainty (a 1,000-sample Monte Carlo over 28 parameters). It is complemented by break-even thresholds for salt deposition, chloride deposition and dew-point rise, and by economic break-even values for CAPEX, land cost and water value. Convergence of the Monte Carlo statistics is verified up to 5,000 samples.
5) An explicit classification of every model component by its evidential basis (Table III) and a field-validation protocol that maps measurements to the parameters they would calibrate (Table XIV).

Scope. The degradation results describe the *electrical degradation of the PV modules*. Degradation of the floating system itself (fatigue of floats and moorings, corrosion of structures [33], [34], storm damage, bio-fouling) is outside the model, so the lifetime energy reported here is not a complete FPV lifetime prediction.

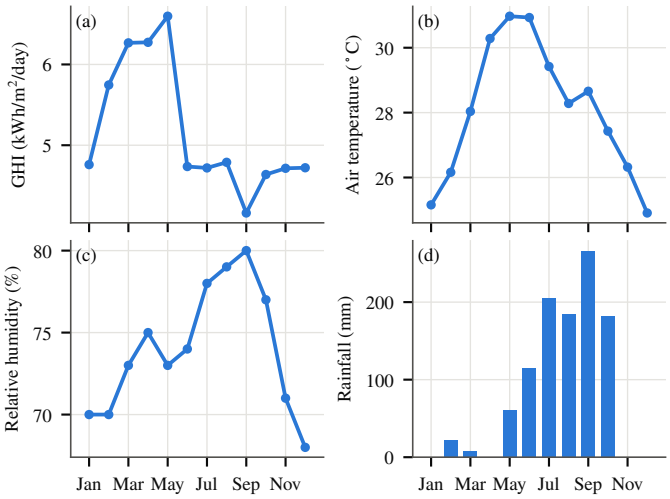


Fig. 1. Monthly climate of the synthetic Visakhapatnam typical year: (a) daily GHI, (b) air temperature, (c) relative humidity and (d) rainfall.

II. SITE AND STUDY SCENARIOS

The reference site is Visakhapatnam (17.69° N, 83.22° E) on the Bay of Bengal. Its climate is tropical, with a hot pre-monsoon season (April–May), the south-west monsoon (June–September) and the north-east monsoon (October–November). An hourly typical year was synthesized from monthly normals (Section III-A). It has a GHI of 1,889 kW h/m^2, a mean air temperature of 28.1 °C, a mean RH of 74.0% and 1,043 mm of rain (Fig. 1).

Three 1-MWp (DC) plants with a DC/AC ratio of 1.2 and monocrystalline PERC modules (−0.35 %/K) are compared (Table II). *Land PV* is a coastal ground-mounted plant on open racks with 15° tilt. *FPV freshwater* is a pontoon plant with 12° tilt on an inland reservoir with low chloride deposition. *FPV saline* uses the same design on near-shore or brackish water (lagoon, salt pan, harbour) with high salt and chloride loading. The baseline values are scenario assumptions informed by the literature, not site measurements; their uncertainty is propagated in Section III-I.

III. MODELING FRAMEWORK

The model chain is shown in Fig. 2. All sub-models run hourly over one year; degradation rates computed from the hourly stresses are projected over a 25-year life. The established irradiance and electrical models follow their reference formulations, as implemented, for example, in pvlib [35]. Table III classifies every component by its evidential basis: established models used as published (A), literature-informed parameter values (B), and modelling assumptions introduced in this work (C). The scientific contribution lies in the coupling and in category C, which is also where field calibration is most needed.

TABLE II
BASELINE SCENARIO PARAMETERS (ASSUMPTIONS)

Parameter	Land	FPV fresh	FPV saline
Tilt (°)	15	12	12
Albedo	0.20	0.07	0.07
U_0 (W/(m^2 K))	25.0	35.0	35.0
U_1 (W s/(m^3 K))	6.84	8.0	8.0
Air–water coupling α	0	0.25	0.30
Dew-point rise ΔT_d (K)	0	1.0	1.5
Wind factor k_v	1.00	1.15	1.25
Dust deposition (g/(m^2 d))	0.15	0.04	0.03
Salt deposition (g/(m^2 d))	0.015	0.010	0.060
D_{Cl} (mg/(m^2 d))	30	8	180
Wiring loss (%)	1.5	2.0	2.0
Wave-motion mismatch (%)	0	0.7	1.0
Availability (%)	99.0	98.5	98.0
CAPEX (INR/Wp)	35	45	52
O&M (INR/kWp/yr)	450	550	750

TABLE III
EVIDENTIAL BASIS OF THE MODEL COMPONENTS

	Components
A	*Established models:* Spencer solar position; Haurwitz clear sky; Erbs decomposition; Hay–Davies transposition; ASHRAE IAM; Kasten–Young air mass; Gueymard precipitable water; First Solar spectral factor; Magnus relation; Faiman module temperature; PVWatts DC and inverter models; Arrhenius/Peck functional form; ISO 9223 concepts of time of wetness and chloride deposition; discounted LCOE.
B	*Literature-informed parameters:* land and FPV heat-loss coefficients and their ranges; Peck E_a and n; PID activation energy; NaCl deliquescence RH (75%); ISO 9223 chloride classes; dust loss coefficient; degradation reference rates calibrated to the magnitude of Indian field surveys.
C	*New modelling assumptions:* over-water microclimate (α, ΔT_d, k_v, T_w); module-surface RH with a 48-h encapsulant lag; dew criterion (1 K/4 K) and dew optical loss; wind-proportional salt deposition and deliquescence cementation; salt-assisted logistic leakage function and threshold shift Δ_s; corrosion scaling $TOW\cdot\sqrt{D_{\mathrm{Cl}}}$; additive mechanisms with a constant annual rate; mitigation factors μ_{PID}, μ_{C}, ϕ and cost adders; all deposition rates and costs.

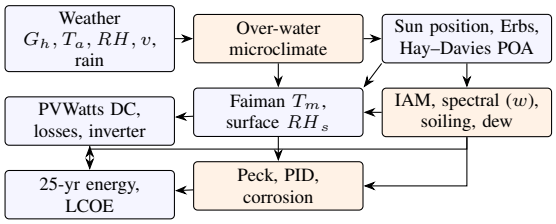


Fig. 2. Coupled model chain. Shaded blocks carry an explicit dependence on humidity or salinity.

A. Weather and Over-Water Microclimate

The synthetic year draws daily rain occurrence and amounts, a first-order autoregressive daily clearness index and daily temperature anomalies from monthly normals. Hourly GHI is the Haurwitz clear-sky value [36] scaled by the clearness index. Air temperature follows an asymmetric diurnal cycle with its minimum near 05:00 and its maximum near 14:00. The dew point T_d is held constant over each day, which reproduces the diurnal swing of RH, and is shifted per month so that the monthly mean RH matches the normal. Wind speed includes a mean-preserving afternoon sea breeze. Table IV compares 20 generated years with the normals they are built from. This shows that the generator is climatologically consistent with its inputs; it is *not* an external validation, because no measured hourly series was available to this study. The RH match is exact by construction, and the monthly clearness index (hence GHI) is a calibrated input that has not been checked against measurements.

TABLE IV
SYNTHETIC-WEATHER CONSISTENCY CHECK: GENERATED YEARS VERSUS INPUT CLIMATOLOGY (20 YEARS)

Variable	Normal	Generated	MBE	RMSE	r
Mean air temperature (°C)	28.3	28.1	−0.24	0.36	0.996
Diurnal range (K)	7.4	7.2	−0.20	0.23	0.993
Mean RH (%)	74.0	74.0	0.00	0.00	1.000
Mean wind speed (m/s)	3.3	3.3	0.01	0.03	0.999
Rainfall (mm)†	1,005.0	944.2	−5.07	9.71	0.995

MBE, RMSE and r are computed over the 12 monthly means. †Annual totals; MBE and RMSE in mm/month.

Over water, the air temperature is pulled towards the water temperature T_w, taken as the 30-day running mean of air temperature plus 0.5 K. The dew point rises by ΔT_d because of evaporation, and the wind speed increases over the smoother surface:

$$T_a' = T_a + \alpha\,(T_w - T_a), \qquad v' = k_v v, \\ T_d' = \min(T_d + \Delta T_d,\, T_a'). \tag{1}$$

Relative humidity follows from the Magnus relation $e_s(T) = 6.112\exp[17.62T/(243.12+T)]$ hPa as $RH = 100\,e_s(T_d')/e_s(T_a')$.

B. Irradiance, Incidence and Spectrum

Solar position uses Spencer's series [37]. GHI is decomposed with the Erbs correlation [38] and transposed to the array plane with the Hay–Davies model [39]:

$$G = G_b\cos\theta + G_d\left[A_i R_b + (1-A_i)\tfrac{1+\cos\beta}{2}\right] + \rho\, G_h \tfrac{1-\cos\beta}{2}, \tag{2}$$

where θ is the angle of incidence, β the tilt, A_i the anisotropy index, R_b the beam ratio and ρ the albedo (0.07 for water, 0.20 for land). Beam reflection losses use the ASHRAE modifier $\mathrm{IAM} = 1 - b_0(1/\cos\theta - 1)$ with $b_0 = 0.05$; diffuse and ground-reflected light take a constant factor of 0.95. Humid air absorbs mainly in infrared bands where c-Si responds weakly, so the mono-Si spectral factor is computed from the Kasten–Young air mass AM [40] and the precipitable water w [41]:

$$M = c_1 + c_2 AM + c_3 w + c_4\sqrt{AM} + c_5\sqrt{w} + c_6\, AM/\sqrt{w}, \tag{3}$$

with the coefficients of [42].

C. Module Temperature, Surface Humidity and Dew

Module temperature follows the Faiman model [43]. A net long-wave loss of $q_{\mathrm{sky}} = 40\,\mathrm{W/m^2}$ to the night sky applies when $G < 50\,\mathrm{W/m^2}$:

$$T_m = T_a' + \frac{G - q_{\mathrm{sky}}\,\mathbb{1}[G<50]}{U_0 + U_1 v'}. \tag{4}$$

The land coefficients are the open-rack defaults. The baseline FPV coefficients (Table II) represent a well-ventilated pontoon

design [7], [8]. Because measured values for other designs are considerably lower [11]–[14], the Monte Carlo ranges include FPV coefficients no better than those of open racks. The RH of the air layer in contact with the glass is

$$RH_{\mathrm{s}} = 100\, e_s(T'_d)/e_s(T_m). \tag{5}$$

Dew forms when $T_m - T'_d < 1\,\mathrm{K}$ and persists after sunrise until $T_m - T'_d > 4\,\mathrm{K}$. While dew is present, an optical loss of 4% is applied. This value is an engineering assumption, not a measured quantity; Section IV-C shows that the conclusions are insensitive to it over 0–6%.

D. Dust and Sea-Salt Soiling

Dust and salt surface densities evolve by a daily mass balance. Dust accumulates at r_d, and salt at $r_s\, v/v_0$ with $v_0 = 3\,\mathrm{m/s}$, because sea-spray generation rises with wind. Rain above 2 mm/day removes a fraction $\eta_d f_\beta$ of the dust and $\eta_s f_\beta$ of the salt, with $\eta_d = 0.80$, $\eta_s = 0.95$ (salt is soluble) and $f_\beta = \sin\beta/\sin 20°$. When the daily maximum RH exceeds the NaCl deliquescence point of 75%, dissolved salt recrystallizes as a crust that cements dust to the glass [29]; on such days the dust-cleaning efficiency is reduced by the cementation factor (baseline 0.5). All plants are cleaned every 15 days with 95% efficiency. The soiling ratio is

$$SR = \max\left(SR_{\min},\, 1 - k_d m_d - k_s m_s\right), \tag{6}$$

with $k_d = 0.035$ and $k_s = 0.060\,\mathrm{m^2/g}$. Two salt quantities are distinguished throughout. The *salt deposition* r_s (g/(m^2 d)) is the mass of sea salt, treated as NaCl, retained on the module glass, which drives soiling and surface leakage. The *chloride deposition* D_{Cl} (mg/(m^2 d)) is the chloride-ion deposition rate of the atmosphere in the ISO 9223 sense [27], which drives corrosion. The deposition rates and coefficients are not site-measured, so they are varied in Sections IV-F and III-I.

E. Electrical Model

The effective irradiance G_{eff} includes the incidence, spectral, soiling and dew corrections. DC power follows the PVWatts model [44]:

$$P_{\mathrm{dc}} = P_0\, \frac{G_{\mathrm{eff}}}{1000}\, [1 + \gamma(T_m - 25)] \prod_j (1 - \ell_j), \tag{7}$$

where $\gamma = -0.0035/\mathrm{K}$ and ℓ_j are the wiring, mismatch, wave-motion and light-induced degradation (LID, 1%) losses. The PVWatts inverter curve (98% nominal) with clipping and an availability factor follow.

F. Module Degradation Model

The annual module degradation rate is the sum of four mechanisms:

$$R = R_0 + R_{\mathrm{TH}} + R_{\mathrm{PID}} + R_{\mathrm{C}}, \tag{8}$$

where $R_0 = 0.30\%$/yr covers ultraviolet exposure and thermal cycling. Encapsulant moisture follows the surface humidity with a first-order lag ($\tau = 48\,\mathrm{h}$) [16]:

$$RH_{\mathrm{e}}(t) = RH_{\mathrm{e}}(t-1) + \kappa\left[RH_{\mathrm{s}}(t) - RH_{\mathrm{e}}(t-1)\right], \tag{9}$$

with $\kappa = 1 - e^{-\Delta t/\tau}$. With the Arrhenius factor

$$a_E(T_m) = \exp\left[-\frac{E}{k_B}\left(\frac{1}{T_m} - \frac{1}{T_{\mathrm{ref}}}\right)\right], \tag{10}$$

the Peck rate [15] averaged over the year is

$$R_{\mathrm{TH}} = R_{\mathrm{TH}}^{\mathrm{ref}} \left\langle a_{E_a}(T_m)\left(\frac{\phi\, RH_{\mathrm{e}}}{RH_{\mathrm{ref}}}\right)^n \right\rangle, \tag{11}$$

with $E_a = 0.79\,\mathrm{eV}$, $n = 3$, $T_{\mathrm{ref}} = 25\,°\mathrm{C}$ and $RH_{\mathrm{ref}} = 60\%$. The factor $\phi = 1$ applies to standard glass–backsheet modules. Because RH_{s} depends on T_m, cooling a module lowers the Arrhenius factor but raises RH_{e}; (11) resolves this trade-off.

PID acts only while the array is energized ($G > 50\,\mathrm{W/m^2}$):

$$R_{\mathrm{PID}} = R_{\mathrm{PID}}^{\mathrm{ref}}(1 - \mu_{\mathrm{PID}})\left\langle a_{E_{\mathrm{PID}}}(T_m)\, f_{\mathrm{leak}}\right\rangle_{\mathrm{on}}, \tag{12}$$

$$f_{\mathrm{leak}} = \left[1 + \exp\left(-\frac{RH - RH^*_{50}}{w}\right)\right]^{-1}, \qquad RH^*_{50} = RH_{50} - \Delta_s\, s, \tag{13}$$

where $E_{\mathrm{PID}} = 0.9\,\mathrm{eV}$, $RH_{50} = 70\%$, $w = 5\%$ RH and $s = \min(m_s/0.5, 1)$ is the salt coverage, with m_s in g/m^2. Two levels of support should be distinguished. *Experimentally supported physics:* PID leakage rises steeply with humidity [20], [21], and sea salt lowers the surface resistance of module glass by orders of magnitude as RH rises and increases leakage current [23], [24]. *Assumed model representation:* the logistic form of (13), its width w, RH_{50} and the threshold shift $\Delta_s = 15\%$ RH for a full salt film are modelling choices that have not been fitted to data. The PID submodel should therefore be interpreted as a physically motivated scenario model rather than an experimentally calibrated predictive PID model.

The corrosion rate follows the ISO 9223 logic of time of wetness and chloride deposition [27]:

$$R_{\mathrm{C}} = R_{\mathrm{C}}^{\mathrm{ref}}\, \frac{TOW}{TOW_{\mathrm{ref}}} \sqrt{\frac{D_{\mathrm{Cl}}}{D_{\mathrm{ref}}}}\, (1 - \mu_{\mathrm{C}}), \tag{14}$$

where TOW counts the hours with $RH > 80\%$ (ambient) or dew on the module, $TOW_{\mathrm{ref}} = 2500\,\mathrm{h}$ and $D_{\mathrm{ref}} = 60\,\mathrm{mg/(m^2\,d)}$.

The reference rates ($R_{\mathrm{TH}}^{\mathrm{ref}} = 0.10$, $R_{\mathrm{PID}}^{\mathrm{ref}} = 0.06$ and $R_{\mathrm{C}}^{\mathrm{ref}} = 0.10\%$/yr) were calibrated so that standard modules on land at this site degrade by roughly 0.9–1%/yr. That is the order of magnitude reported for hot–humid Indian climates [26], above the global median of about 0.5%/yr [25]. They are calibration constants, not measured site values, and are varied over a factor of about two in the Monte Carlo. Likewise, (14) borrows the ISO 9223 variables but not a fitted damage function: the linear time-of-wetness and square-root chloride scaling make it a calibrated scenario representation of corrosion, not a validated corrosion-damage model.

G. Mitigation Options and Their Assumed Effects

Qualification standards define test procedures; they do not establish a numerical reduction of field degradation. The mitigation cases are therefore described by the technology and the standard used to qualify it, and their effect enters the model through *explicitly assumed* factors (Table V). Salt-mist

TABLE V
MITIGATION OPTIONS: QUALIFICATION STANDARDS AND MODEL ASSUMPTIONS

Option	Qualification	Model representation	Assumed value (MC range)
PID-resistant cell/ encapsulant	IEC TS 62804-1:2025	Factor $(1-\mu_{\mathrm{PID}})$ on R_{PID}	$\mu_{\mathrm{PID}} = 0.9$ (0.5–0.95)
Salt-mist-hardened module	IEC 61701:2020	Factor $(1-\mu_{\mathrm{C}})$ on R_{C}	$\mu_{\mathrm{C}} = 0.5$ (0.2–0.7)
Glass–glass, low-permeability encapsulant	No specific standard	Factor ϕ on RH_{e}	$\phi = 0.8$
More frequent cleaning	O&M practice	Cleaning interval	7 days (vs. 15)

Assumed cost adders: 0.5, 1.0 and 2.0 INR/Wp for the three module options and 100 INR/kWp/yr per additional fortnightly cleaning.

resistance is qualified with IEC 61701 [28]. PID susceptibility of crystalline-silicon modules is evaluated with IEC TS 62804-1 [45]; accelerated PID tests have been used to predict field PID rates [46]. The factors μ_{PID} and μ_{C} are sampled over wide ranges in Section III-I. In the rest of the paper, "(M)" denotes modules with both PID-resistant technology and salt-mist hardening.

H. Lifetime Energy and Economics

Year-y energy is $E_y = E_1(1 - R/100)^{y-1}$ over $N = 25$ years. The lifetime projection assumes a constant effective annual degradation rate derived from the stress distribution of the simulated year; the 25-year results therefore represent an equivalent-rate scenario, not a mechanistic year-by-year degradation trajectory. Consequently, the lifetime estimates should be interpreted as equivalent constant-rate projections rather than predictions of the exact annual degradation trajectory. The LCOE is

$$\mathrm{LCOE} = \frac{C_0 + \sum_{y=1}^{N} (O_y - B_y)(1+r)^{-y}}{\sum_{y=1}^{N} E_y (1+r)^{-y}}, \qquad (15)$$

with a discount rate $r = 9\%$ and O&M costs O_y escalating by 5%/yr. $B_y = c_w V_w$ is an optional water benefit: for the freshwater plant, V_w is the evaporation avoided by shading (10 m^2/kWp, 65% suppression of 1800 mm/yr pan evaporation) and c_w a water value. The costs are illustrative Indian-market values, so the economic conclusions are location- and year-dependent. Because LCOE is linear in C_0 and c_w, break-even values follow in closed form.

I. Uncertainty Quantification

Two sources of uncertainty are treated separately. *Weather variability* is represented by 20 independent synthetic years. *Parameter uncertainty* is propagated by Monte Carlo sampling of 28 parameters (1,000 samples): the three degradation reference rates, the dew optical loss (0–6%), the soiling coefficients and cementation factor, the land and FPV heat-loss coefficients (U_0: 22–29 and 22–45 W/(m^2 K); U_1: 4.5–8 and 2.5–10 W s/(m^3 K)), the wind factor, the wave-motion mismatch, α and ΔT_d for each FPV site, the dust, salt and chloride deposition (log-uniform, 0.5×–2×), μ_{PID}, μ_{C}, the CAPEX of each plant, the discount rate (6–12%) and the O&M escalation (2–7%/yr). All are sampled independently from uniform or log-uniform distributions; Table XV in the Appendix lists every parameter with its baseline value, distribution, bounds and justification. Parameters shared by the plants (e.g., the reference rates) take the same draw in all scenarios of a sample, so the scenario differences are paired. Spearman rank correlations identify the dominant parameters. Convergence was assessed by running 5,000 samples and evaluating the statistics on nested subsets of 250–5,000 samples, with bootstrap standard errors; the first 1,000 samples are exactly those reported. The ranges, seeds and baseline values are stored in a machine-readable file in the repository.

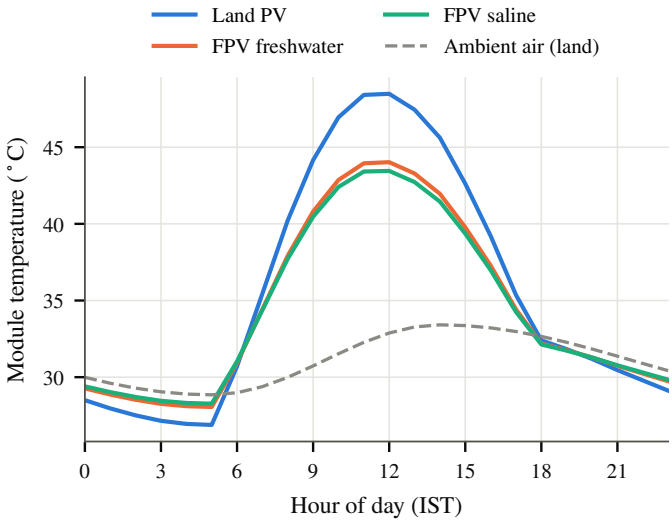


Fig. 3. Mean module temperature by hour of day in May for the three plants, with the site air temperature (baseline parameters).

IV. RESULTS

Unless stated otherwise, the results below are model predictions under the baseline assumptions of Table II.

A. Thermal Behavior

Fig. 3 shows the mean diurnal module temperature in May. The baseline modelled cooling is 3.0 K in daytime over the year, with a peak temperature 7.3 K lower than on land (Table VI). This is a consequence of the assumed FPV heat-loss coefficients, not a measured cooling effect. The daytime module-to-air rise falls from 10.1 K on land to 7.3 K (freshwater) and 7.0 K (saline). At night the air over water stays warmer and the modules follow it, while the higher dew point keeps the air near the array more humid. Dew is present for 1,163, 1,092 and 1,389 h/yr at the land, freshwater and saline plants. The thermal advantage depends directly on the assumed FPV heat-loss coefficients, which are the most uncertain thermal inputs (Section IV-H).

B. Energy Yield and Loss Breakdown

At baseline, freshwater FPV gives the highest year-1 specific yield, 1634 kWh/kWp, +1.6% relative to land (PR 0.837 against 0.819). Table VII and Fig. 4 show where the difference

TABLE VI
YEAR-1 PERFORMANCE (BASELINE)

Metric	Land	FPV fresh	FPV saline
POA irradiation (kW h/m^2)	1963	1953	1953
Specific yield (kWh/kWp)	1608	**1634**	1590
Performance ratio	0.819	**0.837**	0.814
CUF, DC basis (%)	18.35	18.66	18.15
Mean daytime T_m (°C)	39.5	36.5	36.2
Maximum T_m (°C)	57.3	50.0	49.4
Mean $T_m - T_a$, daytime (K)	10.1	7.3	7.0
Mean air RH at array (%)	74.0	77.5	79.7
Dew on glass (h/yr)	1,163	1,092	1,389
Mean daytime soiling ratio	0.965	0.988	0.969

TABLE VII
YEAR-1 LOSS BREAKDOWN (BASELINE). PERCENTAGES ARE SEQUENTIAL LOSSES RELATIVE TO THE PRECEDING STAGE AND ARE NOT DIRECTLY ADDITIVE.

Loss stage (%)	Land	FPV fresh	FPV saline
Incidence angle (IAM)	3.44	3.47	3.47
Spectral (humidity)	-0.23	-0.24	-0.25
Soiling (dust + salt)	3.67	1.27	3.29
Dew / condensation	0.02	0.04	0.05
Module temperature	6.12	4.78	4.64
DC (wiring, mismatch, motion, LID)	3.46	4.62	4.91
Inverter and clipping	2.03	2.03	2.03
Availability	1.00	1.50	2.00

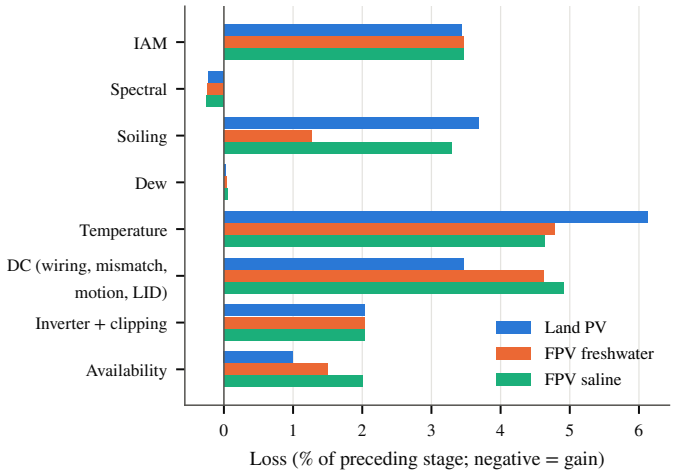


Fig. 4. Year-1 sequential loss breakdown (baseline). Negative values are gains.

comes from. The temperature loss falls from 6.12% to 4.78%, and the soiling loss falls from 3.67% to 1.27% because little dust reaches the water. Part of the gain is lost to higher DC losses (cabling and wave-induced mismatch: 4.62% against 3.46%), lower availability and a slightly lower POA irradiation. Salt soiling brings the saline plant's soiling loss back to 3.29%, and its baseline year-1 yield is 1590 kWh/kWp (-1.1%). Humidity gives a small spectral gain of about 0.24% at all sites, and dew costs less than 0.1%. The absolute soiling losses are scenario estimates, because the salt and dust deposition rates were not measured at the reference site.

The FPV advantage in monthly PR (Fig. 5) is largest in the dry, clear months. The saline plant performs worst in the pre-monsoon months, when strong onshore winds bring salt and

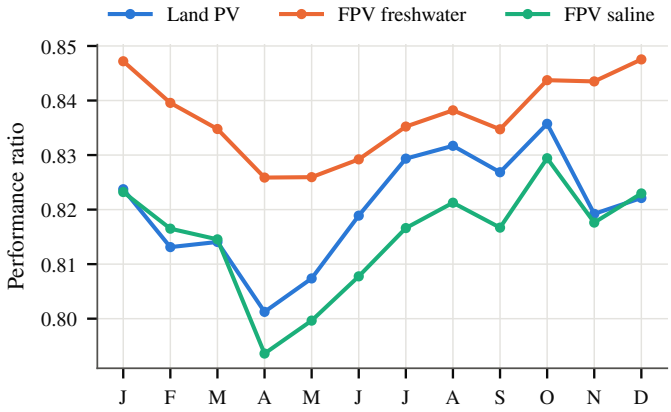


Fig. 5. Monthly performance ratio (baseline).

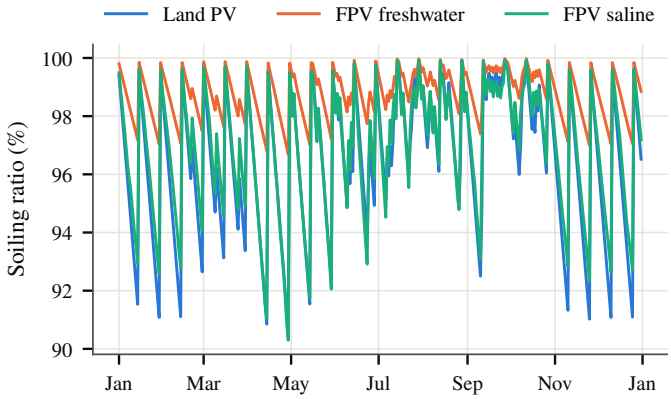


Fig. 6. Daily soiling ratio with rain cleaning, NaCl deliquescence and 15-day manual cleaning (baseline).

little rain falls. Between cleanings, salt deposition gives the saline plant a saw-tooth soiling pattern similar to dust soiling on land (Fig. 6).

C. Sensitivity to Dew Optical Loss

Varying the dew optical loss from 0 to 6% changes the year-1 yield difference relative to land by less than 0.05 percentage point: it ranges from 1.64% to 1.66% for freshwater FPV and from -1.10% to -1.06% for saline FPV. Dew matters through time of wetness, not as an optical loss.

D. Module Degradation

Table VIII and Fig. 7 resolve the modelled degradation by mechanism. Floating modules are cooler, but their encapsulant equilibrates at a higher moisture level: mean RH_{e} is 63.5%, 68.4% and 70.4% at the land, freshwater and saline plants. Through the cubic humidity term of (11), this cancels the Arrhenius benefit of cooling. R_{TH} is 0.34, 0.34 and 0.36%/yr. In this model, therefore, *cooling by water does not translate into slower moisture-driven ageing at a humid site*. PID is predicted at 0.15, 0.15 and 0.21%/yr; the saline value is highest because the salt film makes the glass conductive at lower RH. Corrosion separates the sites most clearly. Time of wetness rises from 3,055 h/yr on land to 3,719 and 4,288 h/yr

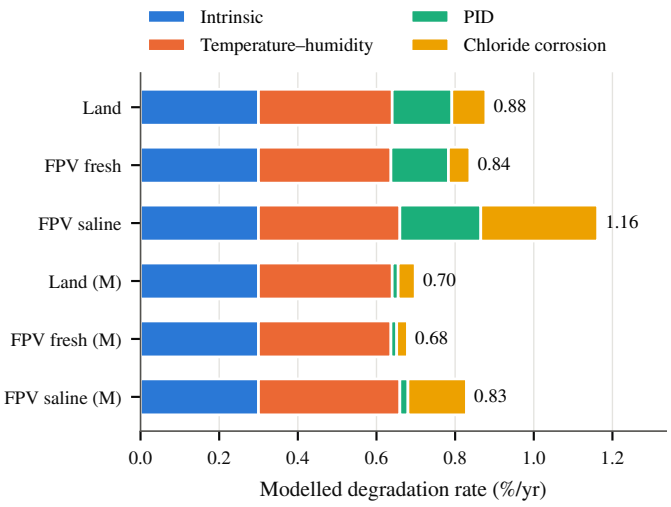


Fig. 7. Modelled module degradation rate by mechanism (baseline). (M): PID-resistant and salt-mist-hardened modules with the assumed factors of Table V.

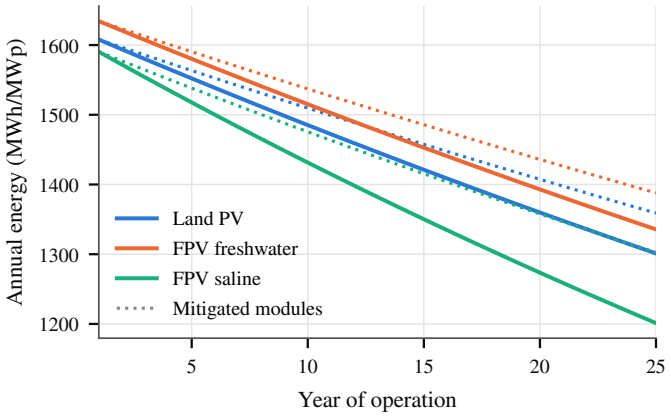


Fig. 8. Annual energy over 25 years (baseline). Dotted lines: PID-resistant and salt-mist-hardened modules (M).

over water, and with 180 mg/(m^2 d) of chloride, R_C reaches 0.30%/yr at the saline plant (0.09%/yr on land). Under the baseline assumptions, total degradation is 0.88, 0.84 and 1.16%/yr.

E. Lifetime Energy and Mitigation

Over 25 years (Fig. 8), the baseline freshwater FPV plant delivers 37,004 MWh/MWp (+2.1% against land) and the baseline saline plant 34,677 MWh/MWp (−4.3%). Table IX evaluates the mitigation options of Table V for the saline plant one at a time. PID-resistant modules alone recover 2.2% of lifetime energy, salt-mist hardening alone 1.7%, and both together 3.9%, still −0.5% below land. Only when a lower-permeability glass–glass package is added does the saline plant exceed the land plant (1.5%); with a 7-day cleaning interval as well, it reaches 3.3%. These are scenario-based mitigation benefits, not validated ones: the recoveries follow directly from the assumed factors and should be read as what-if results.

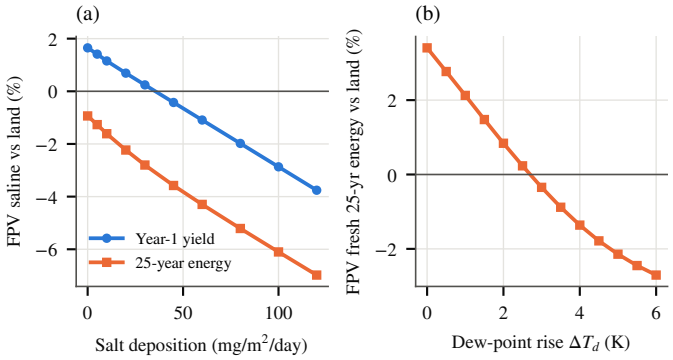


Fig. 9. Break-even thresholds relative to land PV. (a) Saline FPV versus salt-soiling deposition (chloride at baseline). (b) Freshwater FPV 25-year energy versus dew-point rise over water.

F. Thresholds for Salinity and Humidity

Fig. 9(a) varies the salt-soiling deposition of the saline plant with the chloride load held at baseline. Saline FPV outyields land PV in year 1 only below about 35 mg/(m^2 d) of salt deposition. Its 25-year energy stays below land at every salt level, and is still -0.9% at zero salt deposition. Varying the chloride deposition with salt soiling at baseline gives the same picture: the degradation rate rises from 0.87%/yr at zero chloride to 1.31%/yr at 400 mg/(m^2 d), and no chloride level in 0–400 restores lifetime parity (at zero chloride the deficit is -0.9%). Under the baseline assumptions, salt soiling, salt-assisted PID and the extra DC and availability losses each contribute to the saline deficit, so no single mitigation of salinity closes it. For freshwater FPV, Fig. 9(b) varies the dew-point rise over water. The lifetime advantage over land vanishes at $\Delta T_d^* \approx 2.7$ K, where the time of wetness reaches about 5,745 h/yr. This is a threshold predicted by this model under the stated assumptions, not a universal physical constant; it can nevertheless serve as a site-screening criterion. Measuring the humidity increment over a reservoir is therefore a priority for site assessment.

G. Joint Sensitivity to Humidity and Salinity

Fig. 10 varies the dew-point rise over water and scales salt and chloride deposition together for the saline plant. The degradation rate ranges from 0.78 to 1.49%/yr and the 25-year energy from 32,498 to 37,005 MWh/MWp. Each kelvin of dew-point rise adds about 0.167%/yr at baseline salinity. Across the ranges swept, salinity moves lifetime energy by about 2,854 MWh/MWp and moisture by about 1,674 MWh/MWp.

H. Weather Variability Versus Parameter Uncertainty

Weather variability. Over 20 synthetic years (Table X, Fig. 11), the land yield varies between 1,570 and 1,649 kWh/kWp, but the differences between plants are stable, with standard deviations of about 0.1 percentage point. This is stochastic robustness conditional on the weather generator, not empirical validation.

Parameter uncertainty is much larger (Table XI, Fig. 12). Baseline values and Monte Carlo medians answer different

TABLE VIII
MODELLED MODULE DEGRADATION, LIFETIME ENERGY AND LCOE (BASELINE, PER MWP; M = PID-RESISTANT AND SALT-MIST-HARDENED MODULES WITH THE ASSUMED FACTORS OF TABLE V)

	Land	FPV fresh	FPV saline	Land (M)	FPV fresh (M)	FPV saline (M)
R_0 intrinsic (%/yr)	0.30	0.30	0.30	0.30	0.30	0.30
R_{TH} temperature–humidity (%/yr)	0.34	0.34	0.36	0.34	0.34	0.36
R_{PID} (%/yr)	0.15	0.15	0.21	0.02	0.01	0.02
R_{C} chloride corrosion (%/yr)	0.09	0.05	0.30	0.04	0.03	0.15
Total R (%/yr)	**0.88**	**0.84**	**1.16**	**0.70**	**0.68**	**0.83**
Time of wetness (h/yr)	3,055	3,719	4,288	3,055	3,719	4,288
Mean encapsulant RH (%)	63.5	68.4	70.4	63.5	68.4	70.4
Module capacity in year 25 (%)	80.9	81.7	75.5	84.5	84.9	81.9
25-year energy (MWh)	36,233	**37,004**	34,677	37,002	**37,698**	36,045
LCOE (INR/kWh)	2.83	3.54	4.43	2.90	3.60	4.43

TABLE IX
SCENARIO-BASED MITIGATION BENEFITS FOR THE SALINE FPV PLANT (ASSUMED EFFECTS OF TABLE V; PER MWP)

Option	CAPEX (INR/Wp)	R (%/yr)	25-yr energy (MWh)	Recovery vs. standard (%)	vs. land PV (%)	LCOE (INR/kWh)
Standard module	52.0	1.16	34,677	0.0	−4.3	4.43
PID-resistant	52.5	0.98	35,427	2.2	−2.2	4.41
Salt-mist-hardened	53.0	1.01	35,277	1.7	−2.6	4.45
PID + salt-mist	53.5	0.83	36,045	3.9	−0.5	4.43
PID + salt-mist + glass-glass	55.5	0.65	36,792	6.1	1.5	4.50
PID + salt-mist + glass-glass + 7-day cleaning	55.5	0.65	37,421	7.9	3.3	4.54

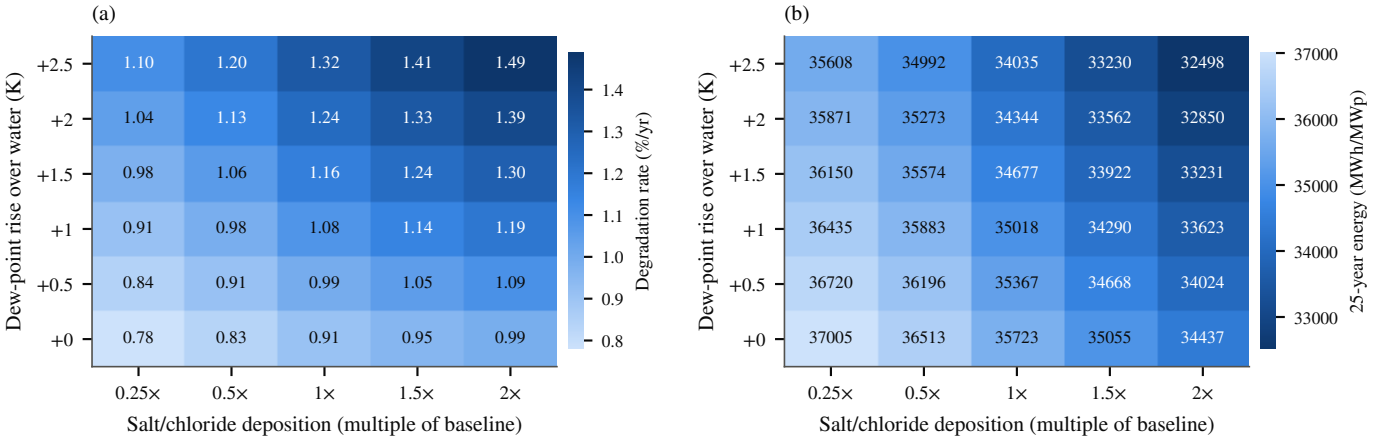


Fig. 10. Saline FPV plant versus dew-point rise over water and salt/chloride load: (a) total modelled degradation rate and (b) 25-year energy.

TABLE X
DIFFERENCES FROM LAND PV OVER 20 WEATHER REALIZATIONS, BASELINE PARAMETERS (MEAN ± SD)

	FPV fresh	FPV saline
Year-1 yield (%)	$+1.71 \pm 0.08$	-1.08 ± 0.05
25-year energy (%)	$+2.17 \pm 0.10$	-4.32 ± 0.06
Degradation rate (%/yr)	-0.039 ± 0.003	$+0.288 \pm 0.002$

questions and are reported separately. For freshwater FPV, the baseline 25-year difference is $+2.1\%$, while the Monte Carlo median is $+1.5\%$ with a 95% interval of $[-3.6, 6.3]\%$, positive in only 75% of samples. For saline FPV, the baseline is -4.3% and the median -5.1% ($[-11.5, 0.7]\%$), negative in all but 4% of samples; with mitigated modules the median is -1.6% ($[-7.0, 3.8]\%$). The medians differ from the baselines because

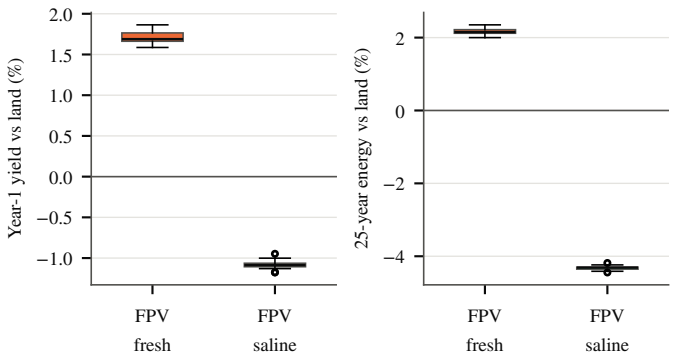


Fig. 11. Weather variability only: distribution over 20 synthetic years of the FPV year-1 yield and 25-year energy relative to land PV.

TABLE XI
PARAMETER UNCERTAINTY: MONTE CARLO OVER 28 PARAMETERS (1,000 SAMPLES, BASELINE WEATHER)

Output	Median	95% interval	$P(>0)$ (%)
Year-1 yield, land (kWh/kWp)	1,604	[1,543, 1,638]	
Year-1 yield, FPV fresh (kWh/kWp)	1,624	[1,589, 1,645]	
Year-1 yield, FPV saline (kWh/kWp)	1,581	[1,520, 1,616]	
Degradation, land (%/yr)	0.89	[0.71, 1.10]	
Degradation, FPV fresh (%/yr)	0.88	[0.68, 1.17]	
Degradation, FPV saline (%/yr)	1.21	[0.91, 1.63]	
Degradation, FPV saline (M) (%/yr)	0.91	[0.67, 1.20]	
Δ yield, fresh $-$ land (%)	1.21	[-1.67, 5.25]	78
Δ yield, saline $-$ land (%)	-1.48	[-5.53, 3.06]	27
Δ 25-yr energy, fresh $-$ land (%)	1.49	[-3.57, 6.33]	75
Δ 25-yr energy, saline $-$ land (%)	-5.08	[-11.50, 0.71]	4
Δ 25-yr energy, saline (M) $-$ land (%)	-1.56	[-6.96, 3.78]	27
LCOE, land (INR/kWh)	2.82	[2.22, 3.58]	
LCOE, FPV fresh (INR/kWh)	3.55	[2.76, 4.55]	
LCOE, FPV saline (INR/kWh)	4.44	[3.52, 5.65]	

TABLE XII
MONTE CARLO CONVERGENCE: 1,000 VERSUS 5,000 SAMPLES. ΔE_{25} AND ΔE_1 ARE THE 25-YEAR ENERGY AND YEAR-1 YIELD RELATIVE TO LAND PV (%); BOOTSTRAP SE OF THE MEDIAN IN BRACKETS

Output	Median		95% interval	
	n=1,000	n=5,000	n=1,000	n=5,000
ΔE_{25}, fresh	1.49 (0.09)	1.42	[-3.57, 6.33]	[-3.55, 6.34]
ΔE_{25}, saline	-5.08 (0.11)	-5.12	[-11.50, 0.71]	[-11.22, 0.45]
ΔE_{25}, saline (M)	-1.56 (0.09)	-1.65	[-6.96, 3.78]	[-7.00, 3.65]
ΔE_1, fresh	1.21 (0.07)	1.11	[-1.67, 5.25]	[-1.74, 5.25]

several ranges are asymmetric about the baseline, notably the FPV heat-loss coefficients, whose range extends to designs with no cooling advantage.

For the saline difference, the rank correlations (Fig. 12b) rank Land dust deposition ($\rho_s = 0.45$), FPV U_1 (0.43) and Saline salt deposition (-0.41) first, followed by FPV U_0 (0.30), Saline chloride deposition (-0.28) and Land U_1 (-0.26). For the freshwater difference the leading parameters are Land dust deposition (0.59), FPV U_1 (0.49) and FPV U_0 (0.35). The baseline freshwater advantage is therefore conditional rather than robust: it rests mainly on the floating design's cooling and on how dusty the land alternative is. The saline penalty, in contrast, persists over most of the plausible parameter space.

Convergence. Table XII and Fig. 13 compare the statistics from the first 1,000 samples with those from all 5,000. Between the two, no median or 95% bound moves by more than 0.28 percentage point, and the bootstrap standard errors of these statistics at 1,000 samples are at most 0.30 percentage point. This is small compared with the width of the intervals, so 1,000 samples are sufficient for the conclusions drawn.

I. Economics and Break-Even Values

Under the assumed Indian-market cost and financing conditions, FPV is more expensive per kWh: 3.54 INR/kWh (freshwater) and 4.43 INR/kWh (saline) against 2.83 INR/kWh on land. In the Monte Carlo, freshwater FPV is more expensive than land PV in 99.5% of samples and saline FPV in 100.0%. Solving (15) for parity with land PV gives the following break-even values:

TABLE XIII
LCOE SENSITIVITY TO FINANCING ASSUMPTIONS (INR/KWH, BASELINE ENERGY)

Case	Land	FPV fresh	FPV saline	Fresh $-$ land	Saline $-$ land
Discount rate 6%	2.34	2.92	3.70	0.58	1.36
Discount rate 9%	2.83	3.54	4.43	0.71	1.60
Discount rate 12%	3.37	4.23	5.24	0.86	1.87
O&M escalation 2%	2.72	3.41	4.24	0.69	1.52
O&M escalation 5%	2.83	3.54	4.43	0.71	1.60
O&M escalation 7%	2.94	3.67	4.61	0.73	1.67

- *CAPEX*: 34.3 INR/Wp for freshwater FPV (baseline 45) and 29.1 INR/Wp for saline FPV (baseline 52).
- *Land cost*: land PV would need a further 10.5 INR/Wp of land-related cost to match freshwater FPV, and 23.6 INR/Wp to match saline FPV.
- *Water value*: valuing the avoided evaporation (11,700 m^3 per MWp and year) lowers the freshwater LCOE to 3.47, 3.39 and 3.24 INR/kWh at 10, 20 and 40 INR/m^3. Parity needs about 93 INR/m^3.

The financing assumptions shift all LCOEs but not their ranking (Table XIII). Over a discount rate of 6–12% and an O&M escalation of 2–7%/yr, the LCOE gap to land PV ranges from 0.58 to 0.86 INR/kWh for freshwater FPV and from 1.36 to 1.87 INR/kWh for saline FPV; both parameters are also sampled in the Monte Carlo. Under these conditions, FPV is justified mainly where land is scarce or costly, where reservoir water has a high value, or where FPV CAPEX falls substantially. No salinity level makes saline FPV cost-competitive at the assumed CAPEX. Because CAPEX, financing, O&M, land and water values vary between locations and over time, these economic results are specific to the stated Indian-market assumptions.

V. DISCUSSION

A. Comparison With Previous Work

The baseline modelled thermal advantage of FPV (about 3 K in daytime, 7 K at peak) lies within the range reported for open, well-ventilated designs [7], [8]. It exceeds the advantage measured for less ventilated systems [11], [14]. The Monte Carlo makes this explicit: with FPV heat-loss coefficients at the low end of the measured range, the freshwater FPV advantage disappears. The predicted salt-driven increase in degradation is larger than the 0.05–0.13 percentage-point effect of salt reported for offshore modules [32]. The difference arises because this work also includes salt soiling, salt-assisted PID and chloride corrosion, whose calibration remains uncertain. Mannino *et al.* [31] found that degradation models need coefficients calibrated to the installation environment; that conclusion applies equally to the reference rates used here.

B. Decision-Relevant Drivers

The rank correlations of Fig. 12(b) show which inputs decide the comparison, and therefore which should be measured first at a candidate site. For freshwater FPV these are the heat-loss coefficients of the floating structure and the dust load at the land alternative. For saline FPV they are the same thermal

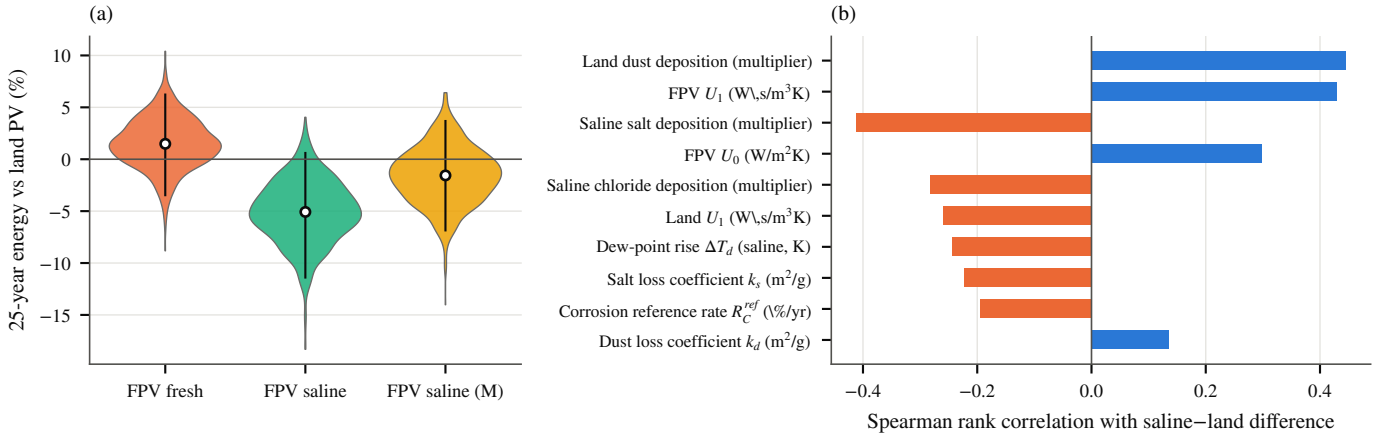


Fig. 12. Parameter uncertainty (1,000 Monte Carlo samples). (a) 25-year energy relative to land PV; dots mark the median and bars the 95% interval. (b) The ten parameters with the largest Spearman rank correlation with the saline minus land difference in 25-year energy.

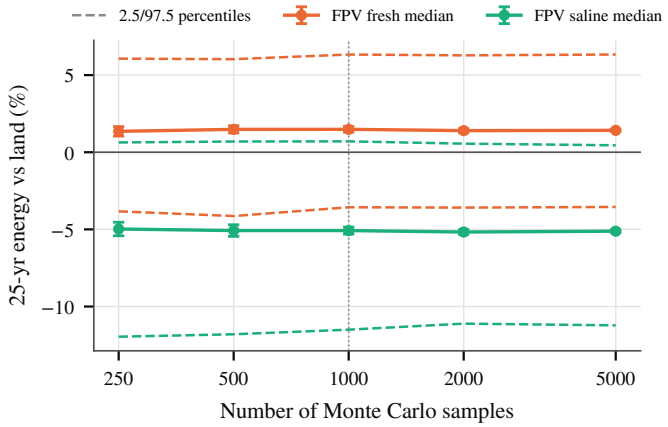


Fig. 13. Convergence of the Monte Carlo median (with ± 1.96 bootstrap SE) and 95% bounds of the 25-year energy difference relative to land PV on nested subsets of one 5,000-sample run. The dotted line marks the 1,000 samples used in the paper.

TABLE XIV
PROPOSED FIELD-VALIDATION MEASUREMENTS AND THE PARAMETERS THEY CALIBRATE

Measurement	Interval	Calibrates
GHI and POA irradiance (pyranometers)	1 min	Decomposition, transposition, clearness
Air temperature, RH, wind over water and over land	1 min	α, ΔT_d, k_v
Water surface temperature	10 min	T_w
Back-of-module temperature (several positions)	1 min	U_0, U_1
Dew/leaf-wetness sensor on module glass	10 min	Dew model, TOW
Soiling reference cells (cleaned/uncleaned)	1 min	r_d, r_s, k_d, k_s, cementation
Chloride deposition (wet candle, ISO 9225)	Monthly	D_{Cl}
Rainfall	1 min	Rain cleaning
DC power of monitored strings	1 min	PR, loss chain
Insulation resistance and leakage current	Daily	f_{leak}, PID onset
IV curves of reference modules	Quarterly	$R_{\mathrm{TH}}^{\mathrm{ref}}$, $R_{\mathrm{PID}}^{\mathrm{ref}}$, $R_{\mathrm{C}}^{\mathrm{ref}}$

and dust inputs together with the salt deposition on the glass and the chloride deposition of the atmosphere. Together with the thresholds of Section IV-F, this gives two practical rules. First, the lifetime benefit of freshwater FPV depends strongly on the thermal characteristics of the floating structure; it should not be assumed from the presence of water alone. Second, the dew-point (humidity) increment over the water should be measured before a lifetime benefit is claimed, because in this model the benefit vanishes above about 2.7 K.

C. Field-Validation Protocol

The framework accepts measured hourly weather and is ready for field calibration. Table XIV proposes a minimum measurement set for an operating FPV plant with a co-located land reference. It maps each measurement to the model parameters it would calibrate. One year of such data would replace the most influential assumptions identified in Fig. 12(b) with site values; multi-year IV-curve and insulation-resistance records would allow the degradation reference rates to be fitted.

D. Implications for Design and Siting

The analysis separates two questions that are often merged: whether FPV produces more energy in year 1, and whether it produces more over its life. For freshwater reservoirs, the baseline answer is yes to both, but the answer depends mainly on the float design's heat-loss coefficients and on the humidity increment over the water. For saline sites, both baseline answers are no, and the lifetime deficit holds over most of the parameter space. These model-based findings suggest the following:

- Verify the heat-loss coefficients of the chosen float design and measure RH and dew point over the water before relying on a lifetime gain.
- At saline or near-shore sites, select PID-resistant module technology (qualified with IEC TS 62804-1) and modules qualified for salt mist (IEC 61701). Consider low-permeability glass–glass packages, and treat the size of their field benefit as uncertain until field data exist.
- Use negative-pole grounding or night-time PID recovery,

marine-grade connectors and corrosion-resistant structures designed for the ISO 9223 corrosivity class of the site.
- At saline sites, shorten the cleaning interval in the dry, windy pre-monsoon season, using demineralized water.
- Monitor module temperature, RH at the array, insulation resistance and soiling in line with IEC 61724-1 [47], so that the parameters can be recalibrated with field data.

E. Limitations

- *Validation.* The weather is a synthetic year that reproduces its input normals (Table IV) but has not been checked against a measured hourly series. No FPV module-temperature or performance data were available to validate the thermal, soiling or degradation models.
- *Assumed parameters.* The over-water microclimate, the deposition rates, the degradation reference rates and the mitigation factors are assumptions. Their uncertainty is propagated, but the ranges themselves are judgements.
- *Degradation physics.* The model is linear in time and additive across mechanisms. It does not represent PID saturation or recovery, interactions between mechanisms, abrupt failures of corroded interconnects, or the higher operating temperature of glass–glass modules.
- *System scope.* Float and mooring fatigue, structural corrosion [33], [34], storm damage, bird soiling and biofouling are not modeled.

VI. Conclusion

A coupled thermal, soiling and module-degradation model was used to compare land-mounted, freshwater floating and saline floating PV for a hot, humid, coastal site in India. With the baseline parameters, the model predicts a baseline modelled cooling of 3.0 K in daytime and gives freshwater FPV a baseline $+1.6\%$ year-1 yield and $+2.1\%$ 25-year energy relative to land. The same water body raises the humidity at the modules, and in the model this cancels the expected temperature benefit for moisture-driven degradation. At the saline site, salt soiling, salt-assisted PID and chloride corrosion raise the predicted degradation rate to 1.16%/yr, and the baseline 25-year energy falls by -4.3%.

Parameter uncertainty is much larger than weather variability, and it changes the interpretation. The Monte Carlo median 25-year difference is $+1.5\%$ for freshwater FPV, with a 95% interval of $[-3.6, 6.3]\%$. The freshwater benefit is therefore conditional: it depends strongly on the thermal characteristics of the floating structure and on the humidity increment over the water, and in this model it vanishes above a dew-point rise of about 2.7 K. For saline FPV the median is -5.1% ($[-11.5, 0.7]\%$), so the model predicts a lifetime penalty across most of the plausible parameter space. Under the baseline parameterization, salinity is the dominant driver of that penalty. Module selection for PID and salt-mist resistance is the most effective modelled mitigation, but its size rests on assumed reduction factors.

These are model-based scenario predictions that still require field validation, for which a measurement protocol is proposed. Assessments of FPV in humid and saline regions should measure the thermal characteristics of the floating structure and the humidity increment over the water, and should report lifetime energy with mechanism-resolved degradation and quantified uncertainty, not first-year yield alone.

Data Availability

The open-source Python implementation (`fpv_analysis`), its unit tests, the scenario parameters and Monte Carlo ranges in machine-readable form (`paper/generated/parameters.json`), all random seeds, the pinned package versions and every generated table are available in the project repository https://github.com/subrahmanyamtanala-rgb/floating-solar-photovoltaic-system-for-humid-and-saline-conditions. The single command `./reproduce.sh` regenerates every number, table and figure in this paper.

Appendix A
Monte Carlo Parameter Space

Table XV lists the 28 parameters sampled in the Monte Carlo. "U" denotes a uniform and "log-U" a log-uniform distribution; multipliers scale the baseline deposition rates of Table II. The same information is stored in `paper/generated/parameters.json`.

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TABLE XV
MONTE CARLO PARAMETERS: BASELINE, DISTRIBUTION, BOUNDS AND JUSTIFICATION

Parameter	Baseline	Distribution	Lower	Upper	Justification
Peck reference rate R_{TH}^{ref} (%/yr)	0.1	U	0.06	0.14	Calibration constant; land total ≈0.9–1%/yr [26]; ±40%
PID reference rate R_{PID}^{ref} (%/yr)	0.06	U	0.03	0.09	Calibration constant; ±50%
Corrosion reference rate R_C^{ref} (%/yr)	0.1	U	0.05	0.15	Calibration constant; ±50%
Dew optical loss	0.04	U	0	0.06	Assumption; effect shown negligible (Sec. IV-C)
Dust loss coefficient k_d (m^2/g)	0.035	U	0.025	0.045	Order of magnitude of dust-soiling studies [48]
Salt loss coefficient k_s (m^2/g)	0.06	U	0.03	0.09	Assumption; no measured salt-crust transmittance
Deliquescence cementation factor	0.5	U	0.25	0.75	Assumption; salt crusts observed on coastal FPV [29]
Land U_0 (W/m^2K)	25	U	22	29	Around open-rack Faiman defaults [43]
Land U_1 (W s/m^3K)	6.84	U	4.5	8	Around open-rack Faiman defaults [43]
Land dust deposition (multiplier)	1	log-U	0.5	2	Assumption; no site measurement
Land chloride deposition (multiplier)	1	log-U	0.5	2	15–60 mg/m^2/day, ISO 9223 class S1 [27]
FPV U_0 (W/m^2K)	35	U	22	45	Measured FPV values [8], [10]–[12]
FPV U_1 (W s/m^3K)	8	U	2.5	10	Measured wind terms [11], [13]
Over-water wind factor k_v	1.15	U	1	1.35	Assumption (lower surface roughness)
Wave-motion mismatch	0.007	U	0.003	0.015	Assumption
Air–water coupling α (fresh)	0.25	U	0.1	0.4	Assumption
Dew-point rise ΔT_d (fresh, K)	1	U	0.5	1.5	Assumption; threshold analysis in Sec. IV-F
Air–water coupling α (saline)	0.3	U	0.15	0.45	Assumption
Dew-point rise ΔT_d (saline, K)	1.5	U	1	2	Assumption
Saline salt deposition (multiplier)	1	log-U	0.5	2	Assumption; threshold analysis in Sec. IV-F
Saline chloride deposition (multiplier)	1	log-U	0.5	2	90–360 mg/m^2/day, ISO 9223 classes S2–S3 [27]
PID mitigation μ_{PID}	0.9	U	0.5	0.95	Assumption (Table V); not implied by IEC TS 62804-1
Salt-mist mitigation μ_C	0.5	U	0.2	0.7	Assumption (Table V); not implied by IEC 61701
Land CAPEX (INR/Wp)	35	U	30	40	Illustrative Indian-market range
FPV fresh CAPEX (INR/Wp)	45	U	38	52	Illustrative Indian-market range
FPV saline CAPEX (INR/Wp)	52	U	44	60	Illustrative Indian-market range
Discount rate	0.09	U	0.06	0.12	Typical range for utility-scale PV finance
O&M escalation	0.05	U	0.02	0.07	Assumption

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